

Knowledge Explorer: An Agentic AI Framework for Interactive, Personalized and Multilingual Learning Experience

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Abstract

Digital education tools have made learning more accessible, yet they often fall short in providing personalized guidance, multilingual support, and engaging pedagogical strategies. As global learners seek more adaptive and inclusive educational experiences, there is a growing need for systems that replicate the explanatory depth of expert tutors. Existing solutions typically lack fine-grained topic breakdowns, contextual grounding, and dynamic delivery methods. To address these gaps, we present *Knowledge Explorer*, a multi-agent AI system that delivers personalized, multilingual explanations using structured topic decomposition, retrieval-augmented generation (RAG), and storytelling. The system leverages LangGraph and LangChain to coordinate agents for subtopic division, factual retrieval, and narrative-based teaching, integrating Cohere's SOTA embedding, rerank and LLM models alongside ChromaDB as the vector store. Our key contributions include a language-agnostic architecture for education, a storytelling-based delivery agent, and a RAG pipeline grounded in authoritative sources. Preliminary results show effective content generation in English, Hindi, and Spanish, underscoring the system's potential as a scalable and globally adaptable educational companion. Overall, these capabilities position Knowledge Explorer as a promising tool for advancing learning equity, particularly in underserved and linguistically diverse communities.

CCS Concepts

• **Information systems** → **Language models**; • **Human-centered computing** → **Human computer interaction (HCI)**; *Interaction design*; • **Social and professional topics** → **K-12 education**;

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Student assessment; • **Computing methodologies** → *Artificial intelligence*; **Natural language processing**.

Keywords

Topic decomposition, Multilingual education, Retrieval-augmented generation, Story-based learning, Personalized learning paths, AI-powered tutoring

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1 Introduction

Education is central to both personal growth and societal advancement, serving as a foundation for knowledge dissemination and skill development across diverse populations. However, many digital learning platforms today largely replicate the shortcomings of traditional education systems by offering generic, one-size-fits-all content that often fails to meet individual learners' unique needs and preferences [12, 14]. Without personalized support, students frequently struggle to engage meaningfully with complex topics, which may hinder their comprehension and reduce motivation. This is particularly true for learners who benefit from structured breakdowns, contextual explanations, or instruction in their native languages [8, 18]. The challenge is amplified in underserved and multilingual communities [17], where access to qualified expert educators and tailored resources is limited or non-existent [1]. As a result, many learners, especially self-directed ones and those with visual or narrative learning preferences, find it difficult to navigate complex educational content effectively.

Despite notable advances in AI-driven educational technologies, significant gaps remain in how these tools address learner diversity and pedagogical effectiveness. Most existing applications lack the capability to systematically decompose topics into manageable subunits that build conceptual understanding progressively, an essential feature for scaffolding learning in a cognitively sound manner [6, 9]. Additionally, these systems often fail to provide clear,

pedagogically grounded explanations or adapt to learners' linguistic backgrounds and cultural contexts [2, 15], which limits accessibility and engagement for a broad user base. This deficiency is particularly problematic in regions where multilingualism is common and educational resources are scarce. Emerging AI techniques such as Retrieval-Augmented Generation (RAG), multi-agent orchestration, and context-aware chat-based frameworks [7, 16] demonstrate promising potential to overcome these challenges. By integrating adaptive content retrieval, narrative generation, and multilingual support, these innovations can enable more inclusive, engaging, and effective learning experiences [5, 11, 13]. However, practical implementations that holistically combine these elements remain limited, motivating further research to bridge these critical gaps.

To address these challenges, we introduce *Knowledge Explorer*, a multi-agent AI platform designed to deliver personalized, multilingual learning experiences. The system combines structured topic breakdown, factual retrieval, and narrative-based teaching to replicate the strategies of expert human tutors. Developed using LangGraph and LangChain, *Knowledge Explorer* integrates Cohere's state-of-the-art multilingual embedding, rerank, and generative models, with ChromaDB as the vector database for semantic retrieval. Our contributions include a language-agnostic architecture for structured education delivery, a storytelling agent to enhance conceptual understanding, and a RAG pipeline grounded in authoritative academic sources [3, 8]. With current support for English, Hindi, and Spanish, *Knowledge Explorer* demonstrates strong potential as a scalable educational assistant capable of improving learning outcomes in diverse and resource-constrained settings.

2 System Design

Our multi-agent architecture enables structured, contextualized, and engaging content delivery through a coordinated pipeline of topic decomposition, retrieval-augmented generation, and multilingual storytelling.

2.1 Workflow

Knowledge Explorer architecture, visualized in Figure 1, presents an AI-powered learning assistant that transforms a user-input topic into an engaging, personalized learning experience. It begins with a Topic Decomposer Agent that breaks the topic into subtopics, followed by a RAG Tool that retrieves relevant content from a Vector Database built from book embeddings. A reranking module (via Cohere API) refines the search results to ensure relevance and quality. The system then uses story generation to explain concepts in a narrative form, enhancing understanding and retention. All processed outputs are compiled and sent to the App UI, delivering an interactive, user-friendly experience. This workflow ensures adaptive, scalable, and engaging education tailored to individual learners.

2.2 Agents

Topic Decomposition Agent. Our Topic Decomposition Agent initiates the workflow by breaking down a user-provided topic into a sequence of non-overlapping subtopics arranged by increasing difficulty. This structured progression—from foundational to

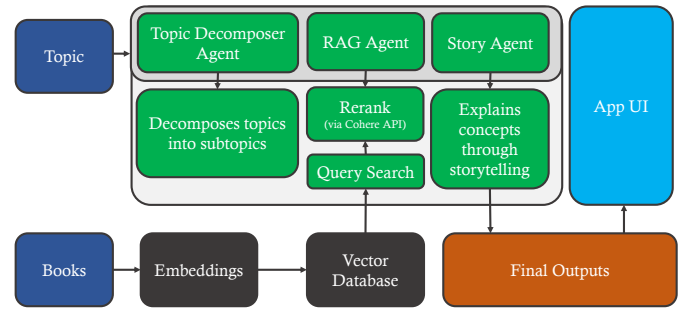


Figure 1: System Architecture of Knowledge Explorer.

advanced—mirrors effective pedagogical practices and supports personalized learning. The agent operates as a three-stage pipeline: a beginner-level node generates elementary subtopics, a middle-level node expands on them with intermediate concepts, and an advanced node introduces specialized themes. At each step, the LLM is guided to avoid redundancy. For example, when given "Force" as input, the agent outputs subtopics like Definition of Force, Newton's Laws, and Continuum Force. This decomposition not only offers clarity on the topic's scope but also enables learners to navigate content based on their current understanding and goals.

RAG Agent. Our RAG Agent in the workflow is responsible for collecting contextually relevant information based on the specific subtopics a user selects. Once a user chooses a subtopic from the output of the Topic Decomposition Agent, the RAG Agent retrieves supporting content from a corpus of over 50 books spanning the domains of physics and biology. We have used the embed-multilingual-light-v3.0 model from Cohere to generate dense vector embeddings for both the subtopic query and the corpus. An initial retrieval step selects the top 20 most relevant documents based on cosine similarity in the embedding space. These 20 candidates are then re-ranked using Cohere's 'rerank-v3.5' model. The top 5 documents from this re-ranking process are selected as the final contextual sources. This two-stage retrieval and re-ranking approach aims to prevent the agent from hallucinating, or going off-topic.

Story Agent. Our Story Agent in the workflow is designed to generate engaging, multilingual narratives based on the contextual information retrieved for each subtopic. Once the RAG Agent has provided the top-ranked context for a user-selected subtopic, the Story Agent takes this information and crafts a coherent and informative story. The agent can generate stories in multiple languages. Each story is tailored to reflect the core ideas of the subtopic, relevant examples, and simplified explanations while maintaining factual accuracy obtained from the RAG.

3 App Features and UI

Knowledge Explorer combines intelligent topic structuring, storytelling-based explanations, and retrieval-augmented content to deliver a personalized and engaging learning experience, as demonstrated in Figure 2.

Guided Topic Selection. The learning experience begins with a simple yet powerful interface where users select a high-level topic, their preferred language, and the subject domain, such as

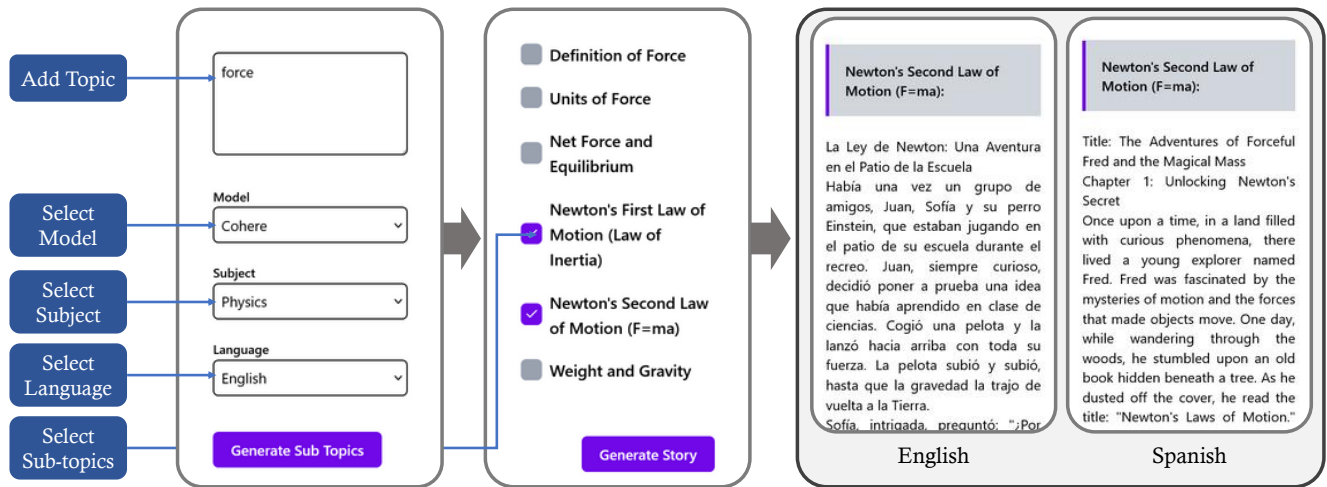


Figure 2: Working Demo of Knowledge Explorer.

Physics or Biology. This initial customization ensures that the app tailors its content generation to the user's interests and linguistic background, fostering a more relevant and accessible learning environment. Once these inputs are provided, the Topic Decomposition Agent is activated, systematically breaking down the chosen topic into a progression of non-overlapping subtopics arranged by increasing difficulty. This structured breakdown not only helps users visualize the entire learning landscape but also guides them through a pedagogically sound learning path—starting with foundational concepts and advancing to more specialized ideas. This clear roadmap empowers users to make informed decisions about their learning journey, whether they wish to build foundational knowledge or explore more complex topics.

Personalized Story Generation. After reviewing the structured list of subtopics, users select the one that aligns best with their curiosity or current knowledge level. Upon selection, the app engages the RAG Agent to retrieve highly relevant contextual information from a carefully curated corpus of over 50 authoritative academic books. This retrieval process uses Cohere's multilingual embedding and re-ranking models to ensure the content is both precise and contextually appropriate. The retrieved context is then passed to the Story Agent, which crafts a compelling, coherent, and educational narrative tailored specifically to the selected subtopic. This storytelling approach transforms dense scientific content into engaging explanations that are easier to comprehend and remember. Importantly, the story is generated in the user's chosen language, supporting deeper understanding across linguistic boundaries and making complex concepts accessible to a broader audience.

Flexible Exploration with Time Travel. A standout feature of Knowledge Explorer is its time travel functionality, designed to enhance user autonomy and curiosity. After reading a generated story for a selected subtopic, users can effortlessly return to the subtopic list to explore other facets of the overarching topic without restarting the entire process. This seamless back-and-forth navigation encourages continuous engagement and exploration, allowing learners to either deepen their understanding by moving

to more advanced subtopics or reinforce earlier concepts by revisiting foundational material. The flexibility supports diverse learning styles and paces, helping users tailor their educational experience to their evolving interests and needs.

Retrieval-Augmented Generation (RAG) for Reliable Content. Knowledge Explorer's content generation is grounded in a robust RAG pipeline that combines advanced retrieval techniques with generative AI, ensuring explanations are accurate, trustworthy, and contextually relevant. By retrieving information from a vetted corpus of academic texts and re-ranking results for precision, the system minimizes the risk of hallucinations or irrelevant content. This approach enhances the reliability of generated narratives, providing learners with factual and pedagogically sound explanations. Furthermore, the multi-stage retrieval and re-ranking mechanism improves response quality and allows the system to dynamically adapt to different topics and query complexities, maintaining high standards of educational integrity.

Multilingual Capability. To address the needs of diverse learners across the globe, Knowledge Explorer is built with multilingual support at its core. The system leverages state-of-the-art multilingual embedding models and generative engines capable of producing high-quality explanations in multiple languages. This capability is crucial for expanding educational access beyond dominant languages, helping bridge gaps faced by multilingual and underserved communities. By delivering content in users' native languages, the app not only improves comprehension but also fosters greater learner confidence and motivation. The language-agnostic architecture is designed to scale easily, allowing additional languages to be incorporated with minimal effort, thereby broadening the app's global reach and inclusivity.

4 User Study and Results

We conducted a comprehensive user study to assess the effectiveness, usability, and adaptability of our *Knowledge Explorer* app in real-world educational scenarios. The study involved 45 participants, including 25 K–12 students and 20 university students

or higher (comprising 15 from STEM, 15 from Arts, and 15 from Business backgrounds), ensuring a diverse range of educational experiences and learning styles. Given the scope of this study, a sample size of 45 participants provides a reasonable balance between diversity and feasibility. It allows us to capture early insights across varied educational levels and disciplines, helping validate core design assumptions before larger-scale deployment.

Questionnaire Design. We began by introducing participants to the motivation, core contributions, key features, demo user interface, and a live deployment of the *Knowledge Explorer* app. Following this orientation, participants were asked to try the app themselves and evaluate the app via a structured questionnaire designed to measure five key aspects of user-centered product design, aligned with the System Usability Scale (SUS) framework [4] which is widely used for evaluating HCI systems [10]. All responses were recorded on a five-point Likert scale, ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). The questionnaire captured feedback across the following dimensions:

- (1) **Ease of Use:** I found the *Knowledge Explorer* interface easy to use.
- (2) **Navigation Confidence:** I felt confident navigating through the educational topics and subtopics.
- (3) **Clarity:** The explanations provided by the AI were clear and understandable.
- (4) **Reusability:** I would like to use this tool frequently for learning new topics.
- (5) **Personalization:** The app adapted well to my learning needs and pace.
- (6) **Engagement:** I found the storytelling and visual elements engaging and helpful.
- (7) **Effectiveness:** I believe this tool could be a good alternative or complement to traditional learning methods.
- (8) **Overall Evaluation:** I thought this intelligent learning assistant was well designed and thoughtfully developed.

Findings. As shown in Figure 3, our AI-powered *Knowledge Explorer* indicate high overall satisfaction (mean = 4.49), with particularly strong scores in Reusability (4.58), Navigation (4.51), and Clarity (4.51), suggesting that users found the system intuitive and easy to understand. Slightly lower scores in Engagement (4.04) and Personalization (4.20) highlight areas for improvement in creating more dynamic and tailored interactions. The low standard deviations across most categories suggest consistent user experiences. These findings support the potential of AI agents as effective, scalable educational tools. We can also notice that university students rated the AI-powered learning tool higher than K-12 students, particularly in Clarity, Reusability, and Engagement, suggesting a better fit for more advanced learners. STEM and Business students also showed high satisfaction, especially in Navigation and Reusability, while Arts students reported lower scores in Personalization and Engagement—indicating a need for more creative and narrative-driven features. Compared to the overall average, feedback from University, STEM, and Business students aligned closely, while slightly lower responses from K-12 and Arts students point to opportunities for further customization.

5 Discussion

Our initial evaluation of *Knowledge Explorer* indicates that the system successfully enhances digital learning through a multi-agent, narrative-driven approach. The structured breakdown of topics into subtopics allowed learners to follow a logical, incremental learning path, which was especially appreciated by secondary and university-level students. The multilingual support—currently in English, Hindi, and Spanish—was praised for improving accessibility and inclusivity, particularly among non-native English speakers. Users consistently highlighted the storytelling feature as both engaging and pedagogically effective, noting that it made abstract or complex topics more relatable. Visual learners benefited from the early stages of AI-generated illustrations, although many suggested improvements in visual clarity and integration with narratives. While overall usability was rated positively, some participants expressed a desire for a more intuitive interface and greater control over the learning journey, such as customizing narrative tone or content depth.

These findings underscore the system’s potential and provide actionable insights for future enhancements, including richer multimodal content and expanded language support. More broadly, *Knowledge Explorer* offers a novel lens for the CSCW community to explore how AI agents can facilitate equitable learning at scale through co-constructive interaction. Its multilingual and agent-based framework opens opportunities for designing culturally grounded educational systems, supporting research on collaborative learning, and bridging global education gaps via adaptive, AI-mediated communication.

6 Concluding Remarks

Knowledge Explorer demonstrates promising potential as an AI-based alternative or supplement to traditional learning methods. By combining structured decomposition, storytelling, and retrieval-augmented content, it offers a scalable solution for diverse learners. While preliminary, the results validate our core design and motivate further development, broader testing, and deeper personalization in future versions. Future directions include integrating adaptive assessment features, enhancing visual explanations using generative models, and expanding multilingual capabilities to support underserved linguistic communities. We also aim to deploy the tool in classroom settings to study long-term learning outcomes and engagement.

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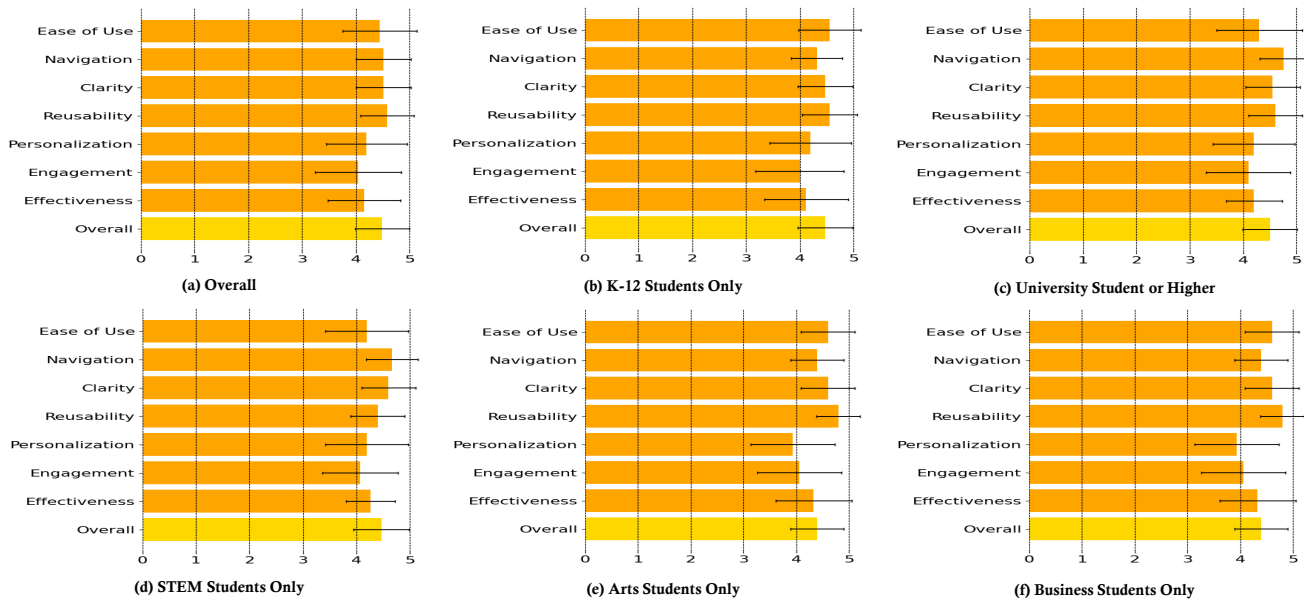


Figure 3: User study results.

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References

- [1] Mohamed Abdelmagied, Mohamed Amine Chatti, Shoeb Joarder, Qurat Ul Ain, and Rawaa Alatrash. 2025. Leveraging Graph Retrieval-Augmented Generation to Support Learners' Understanding of Knowledge Concepts in MOOCs. arXiv:2505.10074 arXiv preprint arXiv:2505.10074.
- [2] Mahfuz Ahmed Anik, Abdur Rahman, Azmine Tushik Wasi, and Md Manjurul Ahsan. 2025. Preserving Cultural Identity with Context-Aware Translation Through Multi-Agent AI Systems. In *Proceedings of the 1st Workshop on Language Models for Underserved Communities (LM4UC 2025)*. 51–60.
- [3] Ahmet Yasin Aytar, Kamer Kaya, and Kemal Kilic. 2024. A Retrieval-Augmented Generation Framework for Academic Literature Navigation in Data Science. arXiv:2412.15404 arXiv preprint arXiv:2412.15404.
- [4] John Brooke. 1995. SUS: A quick and dirty usability scale. *Usability Eval. Ind.* 189 (11 1995).
- [5] Zhendong Chu, Shen Wang, Jian Xie, Tinghui Zhu, Yibo Yan, Jinheng Ye, Aoxiao Zhong, Xuming Hu, Jing Liang, Philip S. Yu, and Qingsong Wen. 2025. LLM Agents for Education: Advances and Applications. arXiv:2503.11733 arXiv preprint arXiv:2503.11733.
- [6] Haoyu Huang, Tong Niu, Rui Yang, and Luping Shi. 2025. RAM2C: A Liberal Arts Educational Chatbot based on Retrieval-augmented Multi-role Multi-expert Collaboration. In *Proceedings of the 31st International Conference on Computational Linguistics*. 448–458.
- [7] Sangwook Lee, Adnan Abbas, Yan Chen, and Sang Won Lee. 2025. CHOIR: Chat-based Helper for Organizational Intelligence Repository. arXiv:2502.15030 <https://arxiv.org/abs/2502.15030>
- [8] Subhankar Maity, Aniket Deroy, and Sudeshna Sarkar. 2025. Leveraging In-Context Learning and Retrieval-Augmented Generation for Automatic Question Generation in Educational Domains. arXiv:2501.17397 arXiv preprint arXiv:2501.17397.
- [9] Elizabeth A. Mullins, Adrian Portillo, Kristalys Ruiz-Rohena, and Aritran Pipalai. 2024. Enhancing Classroom Teaching with LLMs and RAG. In *Proceedings of the 25th Annual Conference on Information Technology Education (SIGITE '24)*.
- [10] Sung-Jun Park, Dong-Kyu Chae, Hong-Kyun Bae, Sumin Park, and Sang-Wook Kim. 2022. Reinforcement learning over sentiment-augmented knowledge graphs towards accurate and explainable recommendation. In *Proceedings of the fifteenth ACM international conference on web search and data mining*. 784–793.
- [11] Devansh Srivastav, Hasan Md Tufsiqur Alam, Afsaneh Asaei, Mahmoud Fazeli, Tanisha Sharma, and Daniel Sonntag. 2025. Enhancing Online Learning Efficiency Through Heterogeneous Resource Integration with a Multi-Agent RAG System. arXiv:2502.03948 arXiv preprint arXiv:2502.03948.
- [12] Jakub Swacha and Michał Gracel. 2025. Retrieval-Augmented Generation (RAG) Chatbots for Education: A Survey of Applications. *Applied Sciences* 15, 8 (2025), 4234.
- [13] Ruotong Wang, Xinyi Zhou, Lin Qiu, Joseph Chee Chang, Jonathan Bragg, and Amy X. Zhang. 2025. Social-RAG: Retrieving from Group Interactions to Socially Ground AI Generation. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. 1–25.
- [14] Shen Wang, Tianlong Xu, Hang Li, Chaoli Zhang, Joleen Liang, Jiliang Tang, Philip S. Yu, and Qingsong Wen. 2024. Large Language Models for Education: A Survey and Outlook. arXiv:2403.18105 arXiv preprint arXiv:2403.18105.
- [15] Azmine Tushik Wasi, Raima Islam, Mst Rafia Islam, Farig Sadeque, Taki Hasan Rafi, and Dong-Kyu Chae. 2025. Dialectal Bias in Bengali: An Evaluation of Multilingual Large Language Models Across Cultural Variations. In *Companion Proceedings of the ACM on Web Conference 2025*. 1380–1384.
- [16] Azmine Tushik Wasi, Taki Hasan Rafi, and Dong-Kyu Chae. 2024. DiaFrame: A Framework for Understanding Bengali Dialects in Human-AI Collaborative Creative Writing Spaces. In *Companion Publication of the 2024 Conference on Computer-Supported Cooperative Work and Social Computing*. 268–274.
- [17] Zheng-Xin Yong, Beyza Ermis, Marzieh Fadaee, Stephen H. Bach, and Julia Kreutzer. 2025. The State of Multilingual LLM Safety Research: From Measuring the Language Gap to Mitigating It. arXiv:2505.24119 <https://arxiv.org/abs/2505.24119>
- [18] Jifan Yu, Zheyuan Zhang, Daniel Zhang-li, Shangqing Tu, Zhanxin Hao, Ruimiao Li, Haoxuan Li, Yuanchun Wang, Hanming Li, Linlu Gong, Jie Cao, Jiayin Lin, Jinchang Zhou, Fei Qin, Haohua Wang, Jianxiao Jiang, Lijun Deng, Yisi Zhan, Chaojun Xiao, Xusheng Dai, Xuan Yan, Nianyi Lin, Nan Zhang, Ruixin Ni, Yang Dang, Lei Hou, Yu Zhang, Xu Han, Manli Li, Juanzi Li, Zhiyuan Liu, Huiqin Liu, and Maosong Sun. 2024. From MOOC to MAIC: Reshaping Online Teaching and Learning through LLM-driven Agents. arXiv:2409.03512 arXiv preprint arXiv:2409.03512.